



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2020– 21
HALF YEARLY EXAMINATION(ONLINE)

CLASS: IX
SUBJECT: MATHEMATICS

FULL MARKS: 40
TIME: 1 HOUR 15 MINUTES

Instructions:

- All questions are compulsory.
- This paper consists of 1 printed page.
- All working including rough work must be clearly shown on the same sheet as the rest of the answers.

Question 1:

- (a) If $x = 7 + 4\sqrt{3}$ then find the value of $x^2 - \frac{1}{x^2}$ [3]
- (b) Factorise : $12mn^2 + 8mn - 20m$ [3]
- (c) A sum of money amounts to ₹11,520 at the end of two years and ₹13824 at the end of three years. Find the sum and the rate of compound interest? [4]

Question 2:

- (a) In what time will ₹2400 amount to ₹2646 at 10% p.a. compounded semi-annually? [3]
- (b) The co-ordinates of points on the x-axis which are at a distance of 5 units from the point (6, -3). Find the points. [3]
- (c) If p and q are rational numbers and $p - q\sqrt{15} = \frac{2\sqrt{3}-\sqrt{5}}{4\sqrt{3}-3\sqrt{5}}$ find the values of p and q. [4]

Question 3:

- (a) Prove that $\sqrt{11}$ is irrational number by the method of contradiction. [3]
- (b) Evaluate:
$$2\sin^2 30^\circ \tan 60^\circ - \frac{3\cos^2 60^\circ}{\cos^2 30^\circ} + \cos^3 90^\circ \tan^2 0^\circ + \tan^2 45^\circ$$
 [3]
- (c) In a parallelogram ABCD, E and F are the midpoints of AB and CD respectively. Show that the line segments AF and EC trisect the diagonal BD. [4]

Question 4:

- (a) Evaluate:
$$\cot 25^\circ \cot 40^\circ \cot 45^\circ \cot 50^\circ \cot 60^\circ \cot 65^\circ$$
 [3]
- (b) Show that the vertices (2,-1), (3,4), (-2,3) and (-3,-2), taken in order, are the vertices of a rhombus. Also find its area. [3]
- (c) if $\sin \theta = \frac{4}{5}$ find the value of $\frac{4\tan \theta - 5\cos \theta}{\sec \theta + \cot \theta}$ [4]